Newton’s Laws of Motion

Name _____

Newton’s first Law of Motion states: A body will continue in its state of motion unless acted upon by an external force.

This means, if a body is at rest, it will remain _____.

If a body is travelling at constant velocity, it will continue _____.

This property of a body to stay in its state of motion is called its **inertia**.

This law explains why a magician can whip a tablecloth out from underneath plates and glasses.

You can try this using a beaker, a coin and a piece of card.

Question 1. Explain why seatbelts help prevent injuries to passengers when a car stops suddenly in a collision.

Question 2. People can get whiplash (neck injury) if the car they are seated in is hit from behind while they are stopped at a traffic light. Explain why, using Newton’s first law.

Question 3. Why is it difficult for a charging elephant to change direction?

Newton's second law of motion describes that a body's acceleration is proportional to the net force acting on it.

This means that a large force will produce a _____ acceleration, while a small force will produce a _____ acceleration.

The mass of the body is also important.

Force = mass x acceleration **or** ***F = m.a***

Force is measured in _____ which has the symbol N.

To calculate force in Newtons, mass must be in kilograms (kg) and acceleration must be in m/s/s (ms^{-2}).

If an object has a large mass, it will need a _____ _____ to accelerate it.

Eg., it needs more force to push a car than to push a bicycle.

Question 4. Describe what would happen to a wheelbarrow's acceleration if two people are pushing it instead of one.

Question 5. If the acceleration caused by gravity is equal to 9.8 ms^{-2}, calculate the force exerted on a falling basketball with a mass of 0.62 kg.

We know that acceleration can be calculated using the equation $a=\frac{v-u}{t}$

So if $F = m \times a$, we can combine the 2 equations to make

$$F=m\frac{(v-u)}{t} \text{ or } \qquad Ft=m(v-u)$$

This equation helps us to understand many applications of Newton's laws – in collisions and in sports.

In a car collision, what this equation means is that if the car stops in a short amount of time, the force acting on the car and its passengers will be high.

That is, if time is small, force will be _____.

This is why many of the road safety devices focus on letting passengers take more time to stop. Crumple zones in cars ensure that the passenger seats keep travelling for a bit longer once the front of the car has hit an obstruction like a wall. Since the car's change in velocity is the same in a collision whether it stops suddenly or not, a l onger time to stop means that the force acting on the passengers is _____, and this reduces the amount of injury.

Question 6. Fences and signposts on the side of the roads are designed so that they give way if they are hit by a vehicle. Explain how this is safer for drivers than the old rigid steel posts and fences.

Question 7. Why are dashboards in cars padded? (You need to mention force and time in your answer.)

Question 8. When a cricketer catches a ball, he will move his hands towards his body in the direction that the ball is travelling. Explain how this allows the cricketer to catch the ball without hurting his hands.

The second application of this equation is to do with sport. Often sportspeople want to hit or kick or throw a ball faster and further. However, the amount of force they can exert is limited by their strength. If the force is allowed to act on the ball for longer, it will cause a larger change in velocity.

This is why tennis players and footballers follow through, keeping the racquet or foot in contact with the ball for a fraction of a second longer.

$$Ft = m(v - u)$$

If time, *t* is increased then the ball's change in velocity will be increased.

Question 9. Explain why most golfers swing the club in a continuous swing, even after the ball has been hit.

Question 10. Many of the top tennis players have their racquets strung with fairly loose strings. How does this enable them to hit a ball at close to 200 kmh^{-1}?

Newton's third law of motion states 'For every force there is an equal and opposite force'.

Eg 1. As you are sitting on your chair, gravity is pulling you downwards. You don't fall to the ground because the chair exerts an equal and opposite force upwards on you.

Eg 2. When we walk, we push back on the ground, and the ground exerts an equal force in the opposite direction, pushing you forward. This process is assisted by friction. If you try to walk on slippery ice, it is difficult to push backward on the ground, so you don't travel forward as easily.

Eg 3. When a rocket launches, it pushes burning fuel down out of the rear of the rocket. The fuel pushes with an equal and opposite force up on the rocket.

Draw a diagram showing this.

Question 11. Use Newton's third law to explain how a swimmer moves through the water.

Question 12. When someone fires a rifle, it recoils into the shooter's shoulder. Explain this using Newton's law.

Choose one of Newton's laws to fill out the table below, then explain your choice.

Sometimes there are more than one correct answer. Your justification is important.

Situation	Newton's law	Why did you choose this law?
We bend our knees when we land after jumping off a table so we don't hurt ourselves		
A rower pulls the oar through the water to propel the boat		
People standing in a bus tend to fall forward when the bus stops		
A basketballer jumps up to the ring		

It takes a long time to stop an ocean liner		
A baseball pitcher follows through with her hand when she throws the ball		
If we are catching an egg, we need to move our hands towards our body so the egg doesn't crack		

Year 10 Physical science assignment

Forces in sport





Choose a sport or other activity (dance, fishing, racing, etc) and research the forces involved in this sport.

You could discuss which forces are helpful in this sport, and which forces are unhelpful. Often friction needs to be reduced in a sport as it causes drag.

Discuss how Newton’s laws of motion apply in this sport.

Your presentation should be approximately 3 pages long. This includes the diagrams that you will provide to illustrate your understandings.

You will be given some time in class to conduct your research.

Due date _____